NCMS: Document Intelligence Pipeline

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Len, I wanted to walk you through what we built over the last three weeks. It started as a cognitive memory system for AI agents. It turned into a full document intelligence pipeline that produces auditable engineering documents from a single prompt.

One prompt. Five documents. Fourteen minutes. Every decision auditable.

What It Does

Type a research topic. Fourteen minutes later, five AI agents have produced: a market research report grounded in 40 sources (web, academic papers, USPTO patents, HackerNews community evidence), a PRD with full traceability, a TypeScript implementation design that passed architecture review at 85%, a requirements manifest, and a structured review report. It works on greenfield research AND existing codebases. Point it at a GitHub repo and the same pipeline produces a modernization plan.

The Memory System: No Vectors Required

Most AI memory systems compress documents into dense vectors and hope cosine similarity finds the right answer. NCMS takes a different approach: three complementary retrieval signals that together beat vector-based systems on real benchmarks.

Signal	What It Does	Why It Matters
BM25 (Tantivy/Rust)	Exact lexical matching	Search 'JWT rotation', get JWT rotation. Not 'auth tokens'.
SPLADE v3	Learned sparse expansion	'API spec' also matches 'endpoint', 'schema', 'contract'.
Graph Activation	Entity relationship traversal	'Connection pooling' finds 'PostgreSQL replication' via shared entities.

nDCG@10 = 0.72 on SciFact, exceeding ColBERTv2 (+4%) and SPLADE++ (+1.5%). 6.3x better temporal reasoning than Mem0. 2.8x better cross-document association than Letta. Zero OpenAI API calls.

The system also tracks how knowledge evolves. **State reconciliation** classifies new facts as supporting, refining, superseding, or conflicting with existing knowledge. **Episodes** cluster related memories via a 7-signal hybrid linker (entity overlap, temporal proximity, domain match, etc.). And **dream cycles** run three offline passes (rehearsal, PMI association learning, importance drift) that teach the system what matters through its own access patterns.

Smart Prompts: Semi-Formal Certificates + Jobs-to-be-Done

We adapted Meta's semi-formal certificate format (arXiv:2603.01896) for every document in the pipeline. The research report must state explicit source premises (S1: "Okta reports 70% workforce MFA adoption"), trace evidence through cross-source analysis with HIGH/MEDIUM/LOW confidence ratings, identify evidence gaps, and derive formal conclusions with citation chains before making recommendations. The PRD must extract research premises (R1-RN) and expert premises (E1-EN), then trace every functional requirement to specific premises. **Result: 29% improvement** in traceability, coverage, and grounding vs standard prompts.

The research also uses a **Jobs-to-be-Done** framework: instead of asking "what features should we build?", it asks "what job is the user hiring this product to do, where are current solutions failing, and where is the market over-serving?" Combined with patent landscape analysis (freedom to operate, coverage gaps) and HackerNews community evidence, the output is a product strategy document, not a literature review.

The NVIDIA Stack

Component	Role
Nemotron 3 Nano	30B params, 256 experts, 3B active. Structured output + reasoning.
DGX Spark	128GB unified memory. 512K context. Sub-second inference. On a desk.
NeMo Agent Toolkit	Agent registration, SSE triggers, FastAPI health checks.

NemoClaw / OpenShell	Kernel-isolated k3s sandboxes. Per-agent network policies.
vLLM (NGC)	Tool calling (qwen3_coder parser), reasoning (nano_v3 parser).

Why the 3B Model Beat the 12B Model

Same pipeline, same prompts, same expert knowledge. Nano passed first try at 85%. Super produced 65KB of broken output (leaked think tags, zero usable content). Qwen needed three rounds and still scored lower. **Model alignment matters more than model size.** Nano was trained for the structured output patterns our pipeline uses. Super couldn't follow the template.

Governance Built In: The Knowledge Workers

Two agents, Architect and Security, don't produce documents. They make everyone else's documents better. Before the pipeline runs, they seed the NCMS memory store with ADRs, CALM model specs, quality attribute scenarios, STRIDE threat models (THR-001 through THR-008), and OWASP ASVS controls. The Product Owner consults them when building the PRD. The Designer consults them again for the implementation. Then both perform structured reviews. **Two agents. Consulted four times.** Every recommendation grounded in governance knowledge. NemoGuardrails enforce policy gates with human-in-the-loop approval when violations are flagged.

The Information Flow Problem

Our first clean run gathered 5 patents in research. The PRD referenced them. The design doc? Zero. Data was lost at every agent handoff. The fix: metadata propagation through the chain, research context in expert questions, and a required "Design Rationale" output section. After: **9 patent citations** with freedom-to-operate assessment and JTBD alignment. The data was always there. The pipeline just needed to carry it.

Document	Baseline	Market Intel	Change
Research	9.4 KB (9 sources)	24.8 KB (40 sources)	+163%
PRD	16.3 KB	24.6 KB	+51%
Design	22.0 KB (0 patent refs)	23.7 KB (9 patent refs)	+7%
Total	**54.4 KB**	**81.4 KB**	**+49%**

Auditable by Design

Every LLM call, document hash, expert consultation, guardrail check, and human approval is persisted with tamper-evident SHA-256 hash chains. JWT-authenticated mutations. Research methodology tracked (planned queries, matched queries, patent abstracts fetched, fallback usage). Compliance scoring: review (40%), guardrails (20%), grounding (15%), approvals (15%), completeness (10%). One-click audit export with chain verification.

What's Next: The Forge

Document Intelligence is Phase 1. The Forge adds: **Code Gen** (auto-implement from design docs, generate PRs), **Test Gen** (test suites from PRD acceptance criteria), **Governance** (automated policy enforcement), **Learning** (dream cycles across projects, pattern recognition). The memory system already supports this. The infrastructure is waiting for the agents.

Five agents. One DGX Spark. One desk. Zero excuses.

Full article: chiefarcheologist.com/blog/ncms-document-intelligence-pipeline

Built with NVIDIA Nemotron Nano, NeMo Agent Toolkit, NemoClaw, LangGraph, and a DGX Spark.